

**A
REPORT
ON**

**IOT-ENABLED SOLAR-ASSISTED HIGH-EFFICIENCY BATTERY
CHARGING USING SUPERCAPACITOR ENERGY BUFFERING**

BY

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**BIRLA INSTITUTE OF TECHNOLOGY &
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**COGNIZANT TECHNOLOGIES SOLUTIONS,
CHENNAI**

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**PREPARED IN PARTIAL FULFILMENT OF THE
WILP DISSERTATION/PROJECT/PROJECT WORK COURSE**

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Key Words: STM32F446RE, ESP8266, IoT Monitoring, PWM Control, Supercapacitor, Embedded Systems, Charging Analysis, Energy Buffering, Cloud Dashboard

Project Areas: Energy Management

Abstract:

The rapid adoption of electric vehicles and portable battery systems has increased the need for efficient and intelligent charging mechanisms in residential environments. Conventional charging systems primarily focus on power electronics for regulating charging voltage and current, but they lack monitoring and adaptive control capabilities for analysing charging behaviour under constrained power conditions.

This project proposes the design and development of an **IoT-enabled smart monitoring and energy-buffered charging system** using embedded controllers and wireless communication modules. The system integrates an STM32 microcontroller for real-time sensing and computation of charging parameters such as voltage, current, power, and energy. A supercapacitor-based buffering mechanism is incorporated to observe charging stability behaviour under fluctuating input conditions.

An ESP8266 module is used as an IoT gateway to transmit sensor data to a cloud-based

dashboard for remote monitoring and analytics. The system enables visualization of charging performance, power consumption trends, and system behaviour in real time.

The prototype demonstrates the feasibility of integrating embedded control with IoT-based monitoring for enhanced visibility of charging systems in constrained power environments.



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LIST OF ABBREVIATIONS:

Abbreviation	Description
ADC	Analog to Digital Converter
CC	Constant Current
CV	Constant Voltage
DC	Direct Current
DC-DC	Direct Current to Direct Current Converter
ESP8266	Wi-Fi Communication Module by Espressif
GND	Ground
I2C	Inter-Integrated Circuit Communication Protocol
IDE	Integrated Development Environment
INA219	Current and Voltage Sensor Module
IoT	Internet of Things
MCU	Microcontroller Unit
PCB	Printed Circuit Board

Abbreviation	Description
PWM	Pulse Width Modulation
UART	Universal Asynchronous Receiver Transmitter
V	Voltage
I	Current
P	Power
E	Energy
Wi-Fi	Wireless Fidelity

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CHAPTER 1 - INTRODUCTION

1.1 Background

The growing adoption of battery-powered devices and electric mobility systems has increased the importance of efficient charging and monitoring mechanisms. Charging systems are commonly used in residential, industrial, and portable electronic applications. In many low-cost charging setups, the charging process operates without providing detailed information about voltage variation, current consumption, power utilization, or charging characteristics.

With the advancement of embedded systems and wireless communication technologies, it has become possible to develop low-cost intelligent monitoring solutions for charging applications. Real-time monitoring enables users to observe charging parameters continuously and analyse the behaviour of the system under different operating conditions.

The Internet of Things (IoT) has further enhanced the capability of embedded systems by enabling remote visualization and cloud-based monitoring of system parameters. IoT-enabled monitoring systems help in collecting, storing, and visualizing data in real time, thereby improving visibility into charging behaviour.

In addition to monitoring, energy buffering techniques using supercapacitors have gained attention due to their high-power density and fast charge-discharge capability. Supercapacitors can temporarily store energy and support short-duration fluctuations in the system. This makes them suitable for studying charging stability and transient energy behaviour.

The proposed project focuses on the design and implementation of an IoT-enabled smart monitoring and analysis system for charging applications. The system integrates sensing, embedded computation, PWM-based control, and wireless communication to observe charging behaviour in real time. A supercapacitor is included as an energy buffering element to study its impact during charging operation.

1.2 Problem Statement

Conventional charging systems primarily focus on supplying power to the load or battery while providing limited visibility into important charging parameters such as voltage, current, power, and energy usage. In many practical scenarios, users are unable to analyse charging behaviour or monitor variations occurring during the charging process.

Existing systems often lack integrated embedded monitoring and cloud-based visualization capabilities. As a result, there is a need for a simple and scalable prototype that combines sensing, embedded computation, and IoT-based monitoring for studying charging characteristics under varying operating conditions.

1.3 Objectives

The main objectives of the project are as follows:

- To design a smart charging monitoring system using embedded technology.
- To measure voltage and current parameters in real time.
- To compute power and energy during charging operation.
- To implement PWM-based control using STM32.

- To transmit charging data to a cloud-based platform using ESP8266.
- To study the effect of supercapacitor buffering on charging behaviour.
- To visualize charging parameters through an IoT dashboard.

1.4 Scope of the Project

The scope of the project is limited to the development of a low-power prototype system for monitoring and analysing charging behaviour. The system includes sensing modules, embedded control, cloud communication, and a supercapacitor-based buffering arrangement.

The project does not involve industrial-scale charging infrastructure or high-voltage EV charging systems. The focus is primarily on embedded monitoring, computation, and analysis of charging parameters in a controlled prototype environment.

1.5 Organization of the Report

This report is organized into multiple chapters describing different aspects of the project work.

- **Chapter 1** presents the introduction, objectives, and scope of the project.
- **Chapter 2** discusses the literature survey related to charging systems, IoT monitoring, and supercapacitor applications.
- **Chapter 3** explains the system architecture and hardware design.
- **Chapter 4** describes the research methodology and computation techniques.

- **Chapter 5** discusses implementation details and module integration.
- **Chapter 6** presents the experimental observations and analysis.
- **Chapter 7** concludes the work and discusses future enhancements.

CHAPTER 2 - LITERATURE REVIEW

2.1 Introduction

The rapid growth of battery-operated systems and electric mobility has increased research interest in charging technologies, energy storage systems, and intelligent monitoring solutions. Several studies have explored the use of embedded systems, IoT platforms, renewable energy integration, and supercapacitor-based energy storage mechanisms for improving charging and energy management applications.

This chapter presents a review of existing work related to charging systems, supercapacitors, embedded monitoring systems, and IoT-enabled energy applications. The review also identifies the research gap that forms the basis for the proposed work.

2.2 Charging Systems and Energy Monitoring

Charging systems are widely used in applications such as portable electronics, battery-operated devices, and electric vehicles. Conventional charging systems are mainly designed to regulate voltage and current supplied to the battery. However, many low-cost systems lack real-time monitoring and analytical capabilities.

Embedded monitoring systems have been increasingly adopted to provide visibility into charging parameters such as voltage, current, power, and energy consumption. Real-time

monitoring allows better understanding of charging behaviour and system performance under varying operating conditions.

Several researchers have focused on integrating microcontrollers and wireless communication technologies into charging systems to enable remote observation and data visualization.

2.3 Supercapacitor-Based Energy Buffering

Supercapacitors are energy storage devices known for their high-power density, fast charge-discharge capability, and long operational life. Unlike conventional batteries, supercapacitors store energy electrostatically rather than chemically.

Research studies have explored the use of supercapacitors as auxiliary energy storage devices in renewable energy systems, electric vehicles, and hybrid storage architectures. Supercapacitors are often used to handle short-duration fluctuations, transient loads, and rapid energy transfer applications.

Hybrid systems combining batteries and supercapacitors have demonstrated improved transient response and enhanced power handling capability. However, most existing research focuses on large-scale systems or advanced industrial applications.

The proposed work uses a supercapacitor primarily as an energy buffering element to observe charging behaviour and short-term stability characteristics in a low-power prototype environment.

2.4 IoT-Based Monitoring Systems

The Internet of Things (IoT) has enabled the development of connected monitoring

systems capable of collecting and transmitting real-time data over wireless networks. IoT platforms are widely used in smart energy systems, industrial automation, and remote monitoring applications.

ESP8266 and similar Wi-Fi modules are commonly used in embedded IoT systems due to their low cost and ease of integration. Cloud platforms such as ThingSpeak and Blynk provide facilities for visualizing sensor data in graphical form.

Several studies have demonstrated the use of IoT for monitoring environmental parameters, energy consumption, and device status. However, many systems focus only on data visualization without integrating embedded computation and charging analysis.

In the proposed system, IoT is used for remote monitoring and visualization of charging parameters rather than direct control of the charging process.

2.5 Embedded Control and PWM Techniques

Embedded controllers are widely used in power electronics and charging applications for sensing, computation, and control. Microcontrollers such as STM32 provide high-speed processing capability and support interfaces required for sensor integration and communication.

PWM (Pulse Width Modulation) is commonly used in DC-DC converters to regulate voltage and current supplied to the load. By varying the duty cycle of the PWM signal, the effective output power can be controlled efficiently.

Several charging systems implement Constant Current (CC) and Constant Voltage (CV) charging techniques using PWM-based control methods. Embedded control also enables implementation of protection mechanisms and monitoring functions.

The proposed work utilizes STM32 for sensor acquisition, embedded computation, and PWM-based regulation of the charging system.

2.6 Research Gap

From the literature survey, it is observed that many existing works focus either on advanced charging infrastructure, renewable energy integration, or industrial-scale energy management systems. Limited work is available on simple and modular prototype systems that combine:

- Embedded sensing and computation
- PWM-based charging control
- IoT-enabled monitoring
- Supercapacitor-based buffering observation

in a single low-cost implementation.

The proposed project attempts to address this gap by developing a practical prototype for monitoring and analysing charging behaviour using embedded systems and IoT technologies.

2.7 Summary

This chapter reviewed existing research related to charging systems, supercapacitors, IoT monitoring, and embedded control techniques. The review highlights the importance of real-time monitoring and embedded computation in charging applications.

Based on the identified research gap, the proposed system focuses on developing a smart

monitoring and analysis platform using STM32 and ESP8266 modules along with a supercapacitor-based buffering arrangement.

CHAPTER 3 - SYSTEM DESIGN AND ARCHITECTURE

3.1 Introduction

The proposed system is designed to monitor and analyse charging behaviour using embedded sensing, computation, and wireless communication techniques. The architecture combines hardware modules responsible for sensing, control, power regulation, communication, and energy buffering.

The system is implemented as a low-power prototype to study charging characteristics in a controlled environment. The overall design focuses on modularity, simplicity, and real-time monitoring capability.

3.2 System Overview

The system consists of the following major modules:

- Power Supply Unit
- DC-DC Converter
- Supercapacitor Buffer
- Battery / Load Module
- Voltage and Current Sensing Unit
- STM32 Embedded Controller

- ESP8266 IoT Communication Module
- Cloud Monitoring Dashboard

The interaction between these modules enables acquisition, processing, and transmission of charging data for real-time monitoring and analysis.

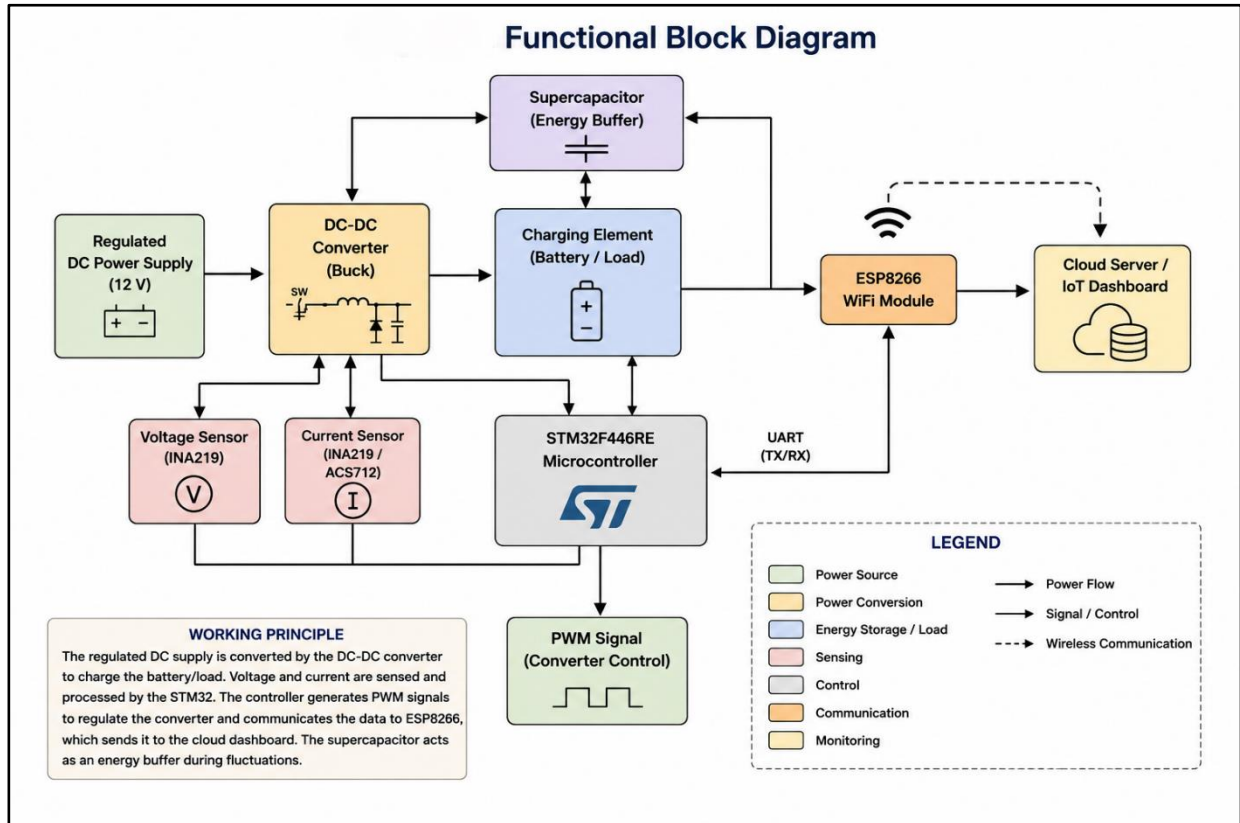


Figure 3.1 Functional Block Diagram

3.3 System Architecture

The proposed system architecture is designed to integrate sensing, embedded computation, power regulation, and IoT-based monitoring into a single modular platform.

The system consists of a regulated power source, DC-DC converter, supercapacitor buffer, sensing unit, STM32F446RE controller, ESP8266 communication module, and a cloud-

based monitoring dashboard.

Voltage and current sensors continuously acquire charging parameters and transmit the measured values to the STM32 controller.

The controller performs real-time computations such as power, energy, and charging analysis while also generating PWM signals for converter regulation.

The processed data is transmitted to the ESP8266 module through UART communication and uploaded to the cloud dashboard for remote visualization and monitoring.

The architecture enables real-time observation of charging behaviour and supports the study of supercapacitor-based energy buffering under controlled operating conditions.

The input power source provides regulated DC power to the charging system. The DC-DC converter regulates the voltage supplied to the battery or load.

A supercapacitor is incorporated as an intermediate energy buffering element to observe charging behaviour under varying conditions.

Voltage and current sensors continuously measure electrical parameters from the system.

The STM32 controller processes sensor data, computes charging parameters, and controls the converter through PWM signals. The processed data is transmitted to the ESP8266 module for cloud-based monitoring.

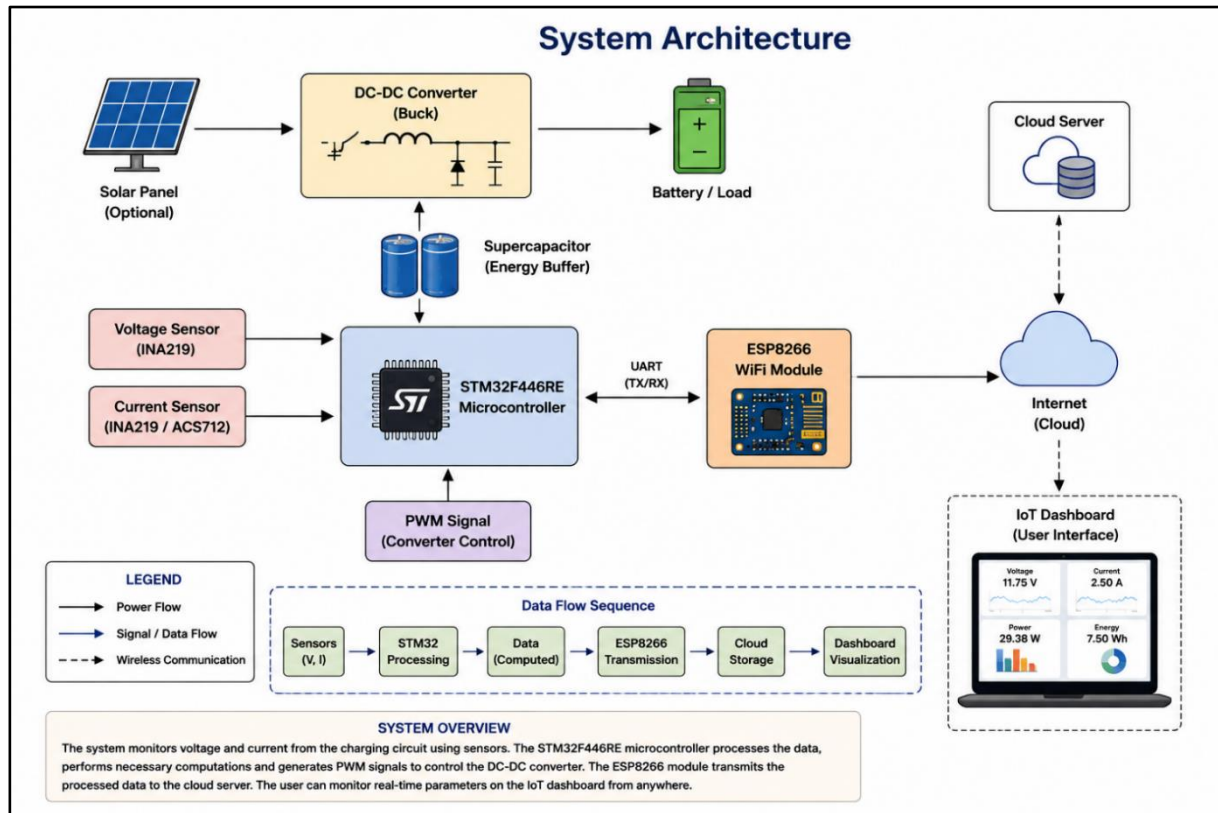


Figure 3.2 Overall System Architecture

3.4 Hardware Modules

3.4.1 STM32F446RE Microcontroller

The STM32F446RE microcontroller acts as the central processing and control unit of the system. It performs the following functions:

- Sensor data acquisition
- Power and energy computation
- PWM signal generation
- Communication with ESP8266

The controller is selected due to its high processing capability, built-in peripherals, and suitability for embedded control applications.

3.4.2 ESP8266 Communication Module

The ESP8266 module provides wireless communication functionality. It receives processed data from the STM32 controller through UART communication and transmits the data to a cloud-based dashboard using Wi-Fi connectivity. The module enables remote visualization of charging parameters such as voltage, current, and power.

3.4.3 Voltage and Current Sensors

Voltage and current measurements are performed using sensor modules connected to the STM32 controller.

The sensing unit performs the following tasks:

- Measurement of charging voltage
- Measurement of charging current
- Data transmission to the controller

These measurements are used for computation and monitoring purposes.

3.4.4 DC-DC Converter

The DC-DC converter regulates the input power supplied to the battery or load. The converter operates based on PWM control signals generated by the STM32 controller. The converter enables controlled charging operation by adjusting voltage and current levels.

3.4.5 Supercapacitor Buffer

A supercapacitor is used as a temporary energy storage element within the charging system. Due to its fast charge and discharge capability, the supercapacitor helps in handling short-duration fluctuations and transient variations. The supercapacitor is incorporated primarily for observation and analysis of buffering behaviour during charging.

3.4.6 Battery / Load Module

The battery or equivalent load acts as the charging element in the prototype system. The charging behaviour of the battery is monitored using embedded sensing and cloud visualization.

3.4.7 Power Supply Unit

A regulated DC power source is used to supply energy to the prototype system. The supply voltage is maintained within safe operating limits to ensure stable operation of all hardware modules.

3.4.8 Hardware Setup Implementation

The hardware setup diagram illustrates the physical arrangement and interconnection of the major components used in the proposed smart charging monitoring system. The setup includes the STM32F446RE controller, ESP8266 communication module, voltage and current sensing units, DC-DC converter, supercapacitor buffer, and regulated power supply.

The STM32 controller performs real-time sensing, computation, and PWM-based control,

while the ESP8266 module enables wireless communication with the cloud dashboard. Proper grounding, stable power connections, and reliable sensor interfacing are maintained to ensure accurate measurement and stable operation of the system. The hardware arrangement is designed using a modular approach, allowing easy testing, debugging, and future expansion of the prototype system.

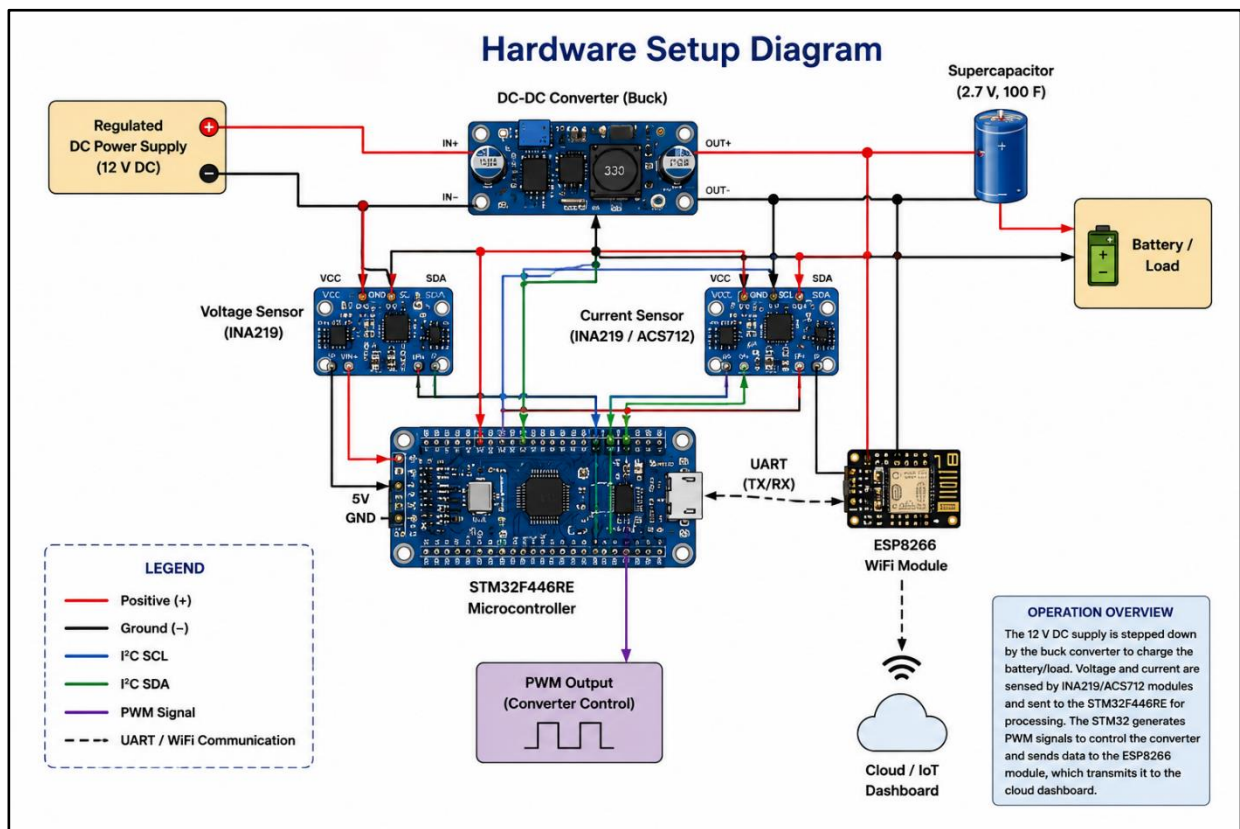


Figure 3.3 Hardware Set up Diagram of Proposed System

3.4.9 Hardware Circuit Diagram

The hardware circuit diagram represents the electrical interconnection between the major components of the proposed charging monitoring system. It includes the regulated power supply, DC-DC converter, supercapacitor buffer, sensing modules, STM32F446RE controller, and ESP8266 communication module.

The circuit is designed to enable real-time sensing, embedded computation, PWM-based converter control, and wireless transmission of charging parameters. Proper grounding and signal interfacing are maintained to ensure stable operation and accurate measurement of voltage and current values during system operation.

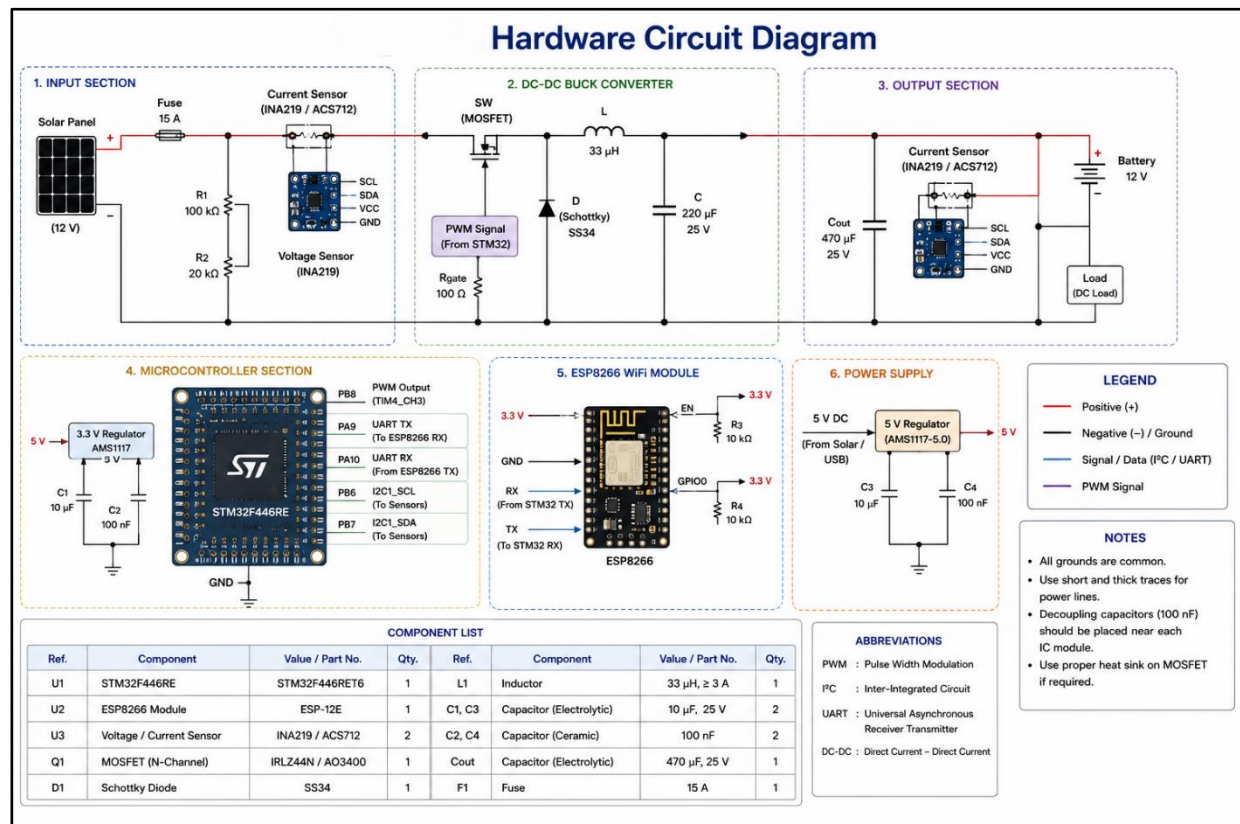


Figure 3.4 Hardware Circuit Diagram of Proposed System

3.5 Communication Interfaces

3.5.1 UART Communication

UART communication is used between STM32 and ESP8266 modules for serial data transmission. Sensor data and computed values are transmitted in serial format.

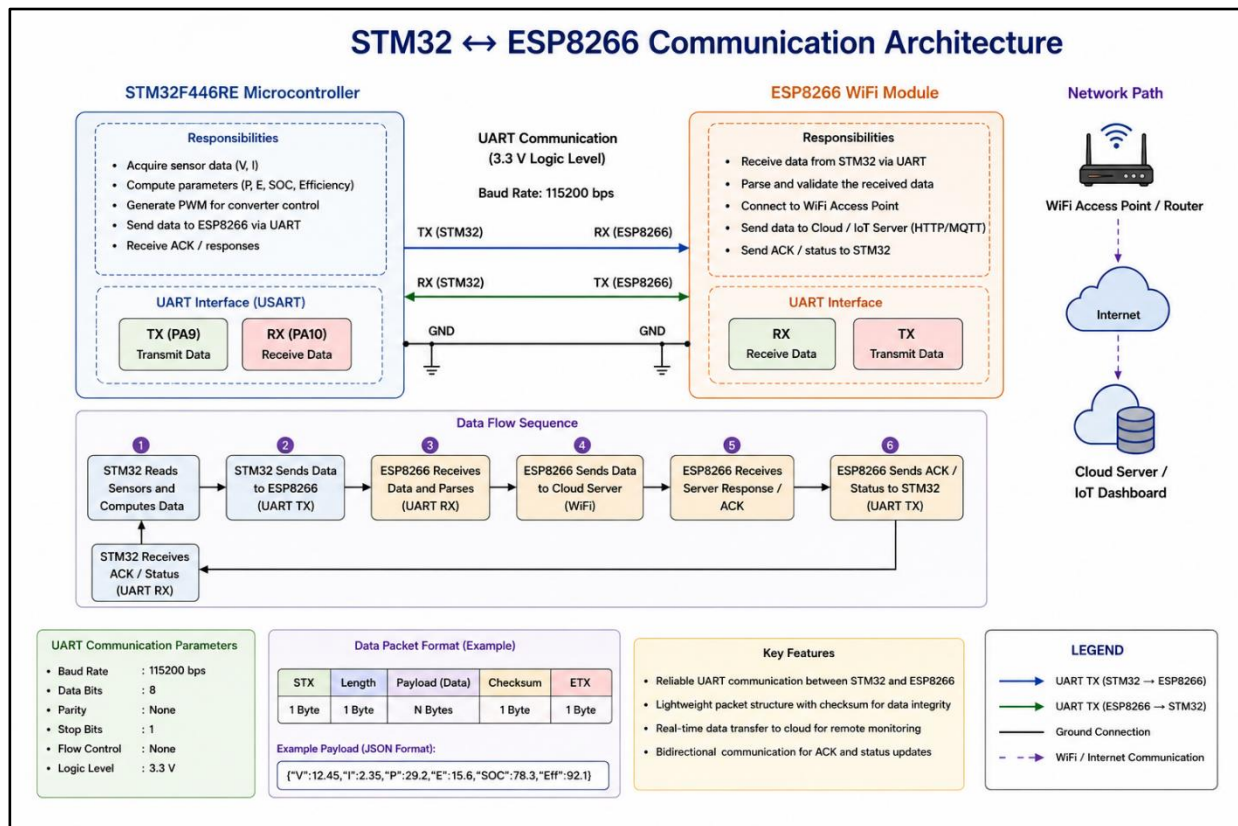


Figure 3.5 STM32 and ESP8266 Communication Architecture

3.5.2 PWM Interface

PWM signals generated by STM32 are used to control the operation of the DC-DC converter. The duty cycle of the PWM signal determines the effective output voltage and current.

3.5.3 Sensor Interface

Voltage and current sensors are connected to STM32 through analog or digital interfaces. The controller continuously reads sensor values during operation.

3.5.4 Wi-Fi Communication

The ESP8266 module connects to a wireless network and transmits charging data to the cloud dashboard for visualization and monitoring.

3.6 System Operation

The system operates in the following sequence:

1. The regulated DC supply provides power to the charging system.
2. The DC-DC converter regulates voltage supplied to the load.
3. Voltage and current sensors measure charging parameters.
4. STM32 acquires sensor data and performs computations.
5. PWM signals are generated for converter control.
6. Computed data is transmitted to ESP8266 through UART communication.
7. ESP8266 uploads the data to the cloud dashboard.
8. Charging behaviour is monitored and analysed in real time.

3.7 Design Considerations

The system is designed based on the following considerations:

- Low-cost implementation
- Modular hardware structure
- Real-time monitoring capability
- Safe operating voltage levels
- Ease of integration and testing

- Wireless communication support
- Simple and scalable architecture

3.8 Summary

This chapter described the system architecture, hardware modules, interfaces, and operational flow of the proposed charging monitoring system. The architecture combines embedded control, sensing, communication, and energy buffering techniques to enable real-time monitoring and analysis of charging behaviour.

CHAPTER 4 - RESEARCH METHODOLOGY

4.1 Introduction

This chapter describes the methodology adopted for the development of the proposed smart charging monitoring system. The methodology includes system design, hardware integration, embedded programming, data acquisition, computation, communication, and monitoring. The overall approach focuses on implementing a modular prototype capable of measuring and analysing charging parameters in real time using embedded systems and IoT technologies.

4.2 Methodology Overview

The development methodology is divided into the following stages:

1. Problem Identification
2. System Design
3. Hardware Selection

4. Sensor Integration
5. Embedded Programming
6. Communication Setup
7. Cloud Monitoring
8. Testing and Analysis

Each stage contributes to the implementation of the complete monitoring system.

4.3 Problem Identification

Many low-cost charging systems provide limited visibility into charging behaviour and electrical parameters. Users are often unable to monitor voltage, current, power consumption, and charging variations in real time. The project aims to address this limitation by integrating sensing, embedded computation, and IoT-based monitoring into a single prototype platform.

4.4 System Design Methodology

The system architecture is designed using a modular approach to simplify integration and testing.

The design methodology includes:

- Selection of a suitable embedded controller
- Identification of sensing requirements
- Integration of communication modules

- Implementation of power regulation
- Inclusion of a supercapacitor buffering arrangement

The modular design allows independent testing of individual modules before complete integration.

4.5 Hardware Selection

The hardware components are selected based on simplicity, cost-effectiveness, and compatibility with embedded applications.

4.5.1 STM32F446RE Controller

The STM32F446RE microcontroller is selected as the main processing unit due to its:

- High processing speed
- PWM generation capability
- Multiple communication interfaces
- ADC support for sensor acquisition

4.5.2 ESP8266 Module

ESP8266 is selected for wireless communication because of:

- Built-in Wi-Fi support
- Ease of integration
- Low cost

- Compatibility with cloud platforms

4.5.3 Sensors

Voltage and current sensors are selected to continuously monitor charging parameters.

The sensing unit enables:

- Voltage measurement
- Current measurement
- Data acquisition for computation

4.5.4 Supercapacitor

A supercapacitor is included as an energy buffering element to study its behaviour during charging operation. Its high-power density and fast charge-discharge capability make it suitable for observing short-duration fluctuations.

4.6 Data Acquisition Methodology

Voltage and current data are acquired continuously using sensors connected to the STM32 controller.

The acquired data is:

- Sampled periodically
- Converted into digital values
- Processed for further computation

The sensor data forms the basis for power and energy calculations.

4.7 Embedded Computation

The STM32 controller performs real-time computations using sensor data acquired from the charging system. These computations are used for monitoring, analysis, and PWM-based control.

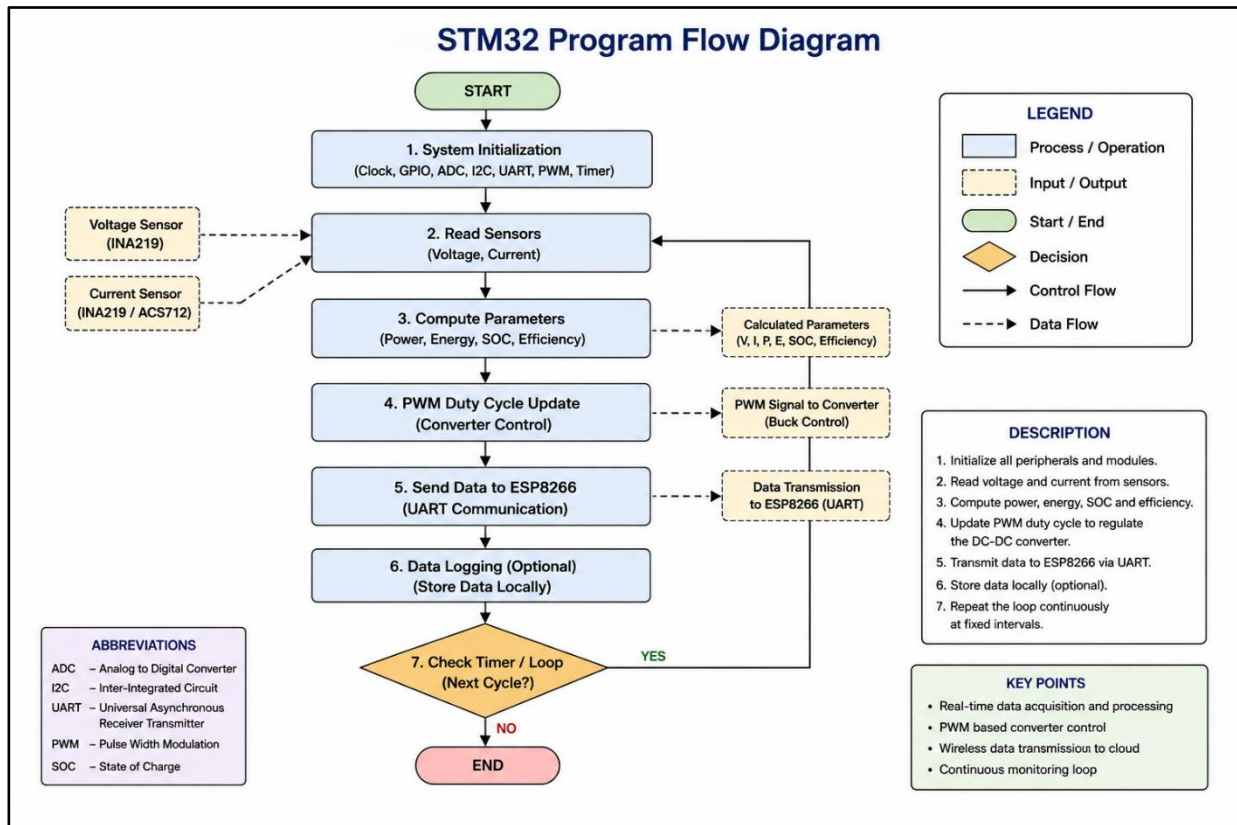


Figure 4.1 STM32 Program Flow for Charging Computation

4.7.1 Power Calculation

The instantaneous power delivered to the charging system is calculated using:

$$P = V \times I$$

Where:

- P = Power (Watts)
- V = Voltage (Volts)
- I = Current (Amperes)

The STM32 controller continuously acquires voltage and current data through sensors and computes power in real time.

4.7.2 Energy Calculation

The total energy transferred during charging is estimated by integrating power over time.

$$E = \int P dt$$

For discrete embedded implementation:

$$E = P \times t$$

Where:

- E = Energy
- P = Power
- t = Time duration

The cumulative energy value is used for charging analysis and system monitoring.

4.7.3 Supercapacitor Energy Estimation

The energy stored in the supercapacitor is estimated using:

$$E = \frac{1}{2} CV^2$$

Where:

- C = Capacitance (Farads)
- V = Voltage across supercapacitor

This computation helps in estimating the buffering capability of the supercapacitor during transient conditions.

4.7.4 PWM Duty Cycle Calculation

PWM duty cycle is calculated using:

$$D = \frac{V_{out}}{V_{In}} \times 100$$

Where:

- D = Duty Cycle (%)
- V_{out} = Required output voltage
- V_{In} = Input voltage

The STM32 dynamically adjusts PWM duty cycle to regulate converter operation.

4.7.5 State of Charge Estimation

Battery State of Charge (SOC) is estimated using:

$$SOC = \frac{Q_{charged}}{Q_{max}} \times 100$$

Where:

- $Q_{charged}$ = Current charge level
- Q_{max} = Maximum battery capacity

SOC estimation helps in monitoring charging progress and charging state transitions.

4.8 PWM-Based Control Methodology

PWM (Pulse Width Modulation) is used to regulate the operation of the DC-DC converter.

The STM32 controller generates PWM signals with varying duty cycles to:

- Regulate output voltage
- Control charging current
- Maintain stable operation

Duty cycle adjustment enables efficient control of power delivered to the load.

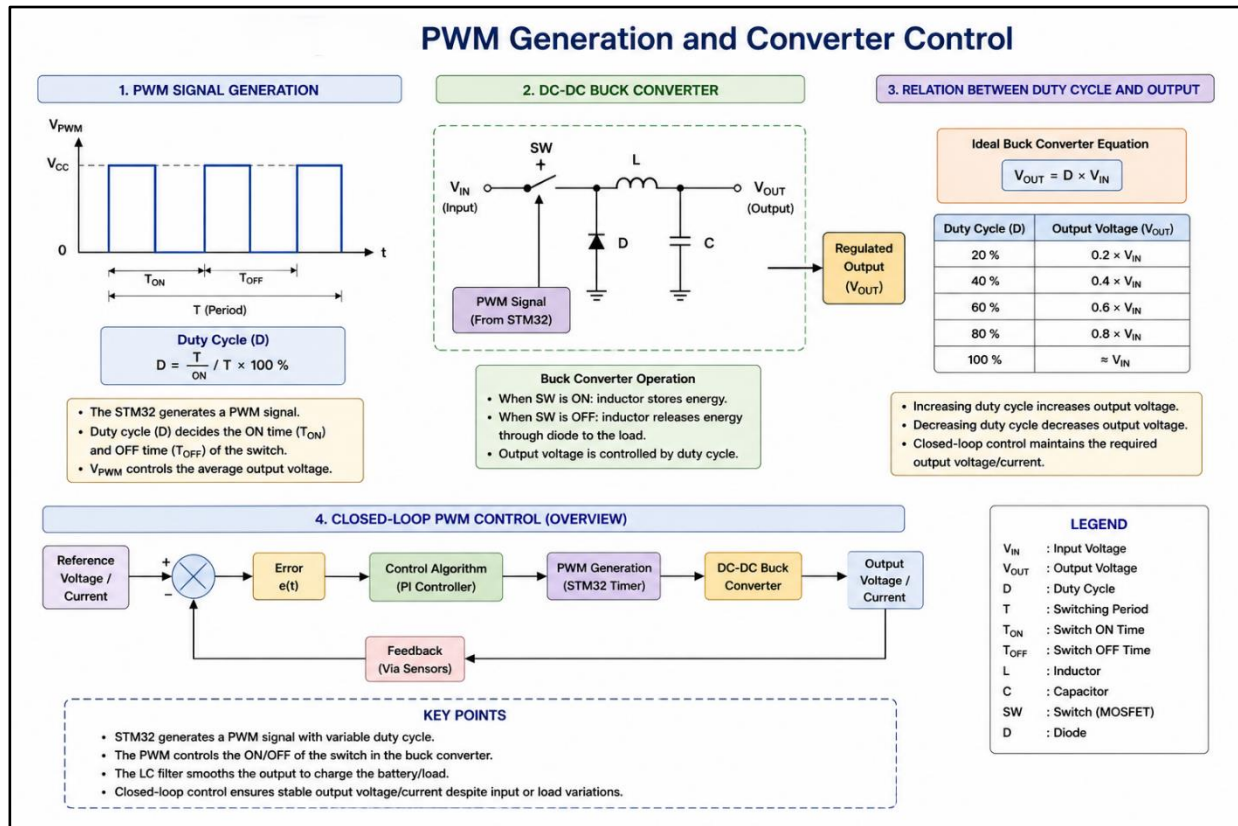


Figure 4.2 PWM Generation and DC-DC Converter Control

4.9 Communication Methodology

Communication between STM32 and ESP8266 is established using UART serial communication.

The communication process includes:

1. Sensor data acquisition
2. Data formatting by STM32
3. Serial transmission to ESP8266
4. Wi-Fi transmission to cloud platform

This enables remote monitoring and visualization of charging parameters.

4.10 Cloud Monitoring Methodology

The cloud platform is used for visualizing charging data in graphical form.

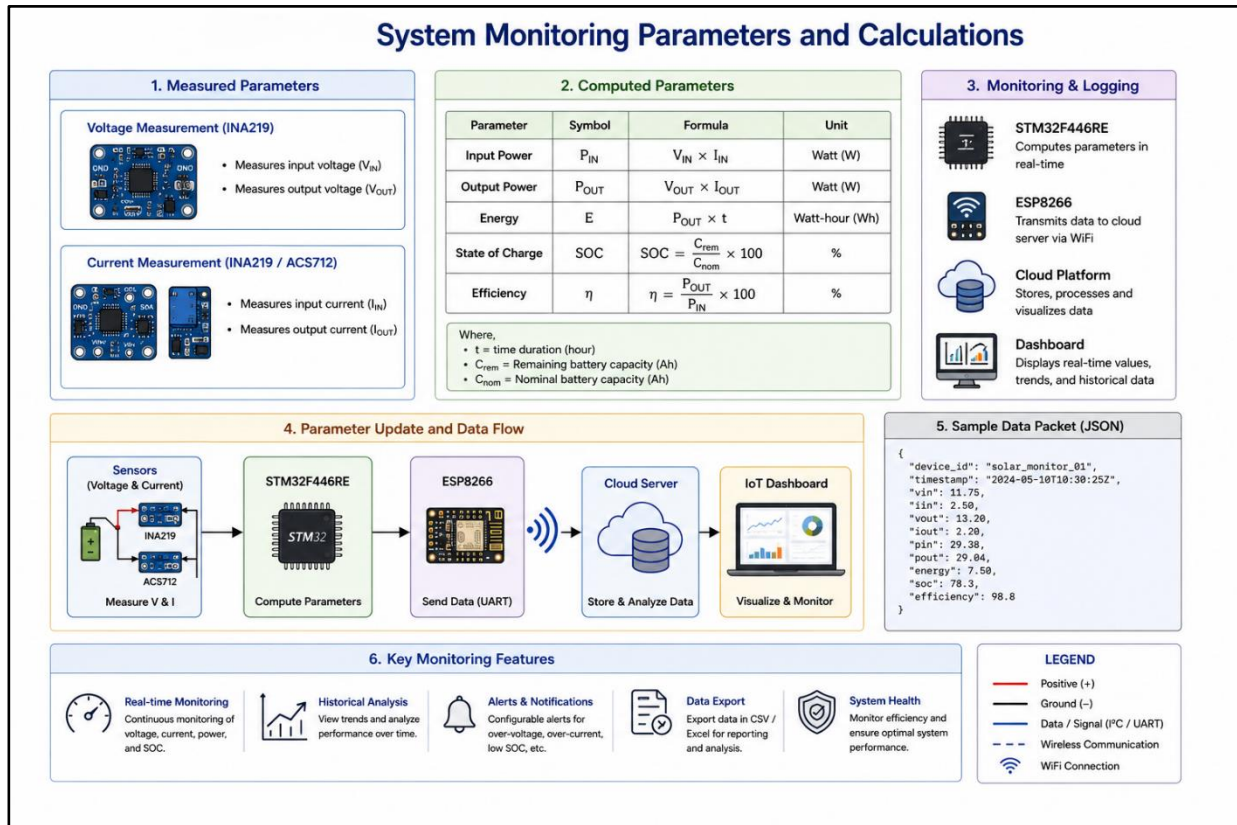


Figure 4.3 Monitoring Parameters and Embedded Calculations

The monitoring dashboard displays:

- Voltage variation
- Current variation
- Power values
- Charging trends over time

Cloud-based monitoring enables observation and analysis of system behaviour remotely.

4.11 Experimental Methodology

The experimental setup is developed to observe charging behaviour under controlled conditions.

The testing procedure includes:

- Operation of the charging system under regulated DC supply
- Monitoring voltage and current values
- Observation of supercapacitor behaviour
- Visualization of charging parameters through the dashboard

Comparative observations may also be performed with and without the supercapacitor buffering arrangement.

4.12 Firmware Execution Flow

The firmware execution flow diagram illustrates the sequence of operations performed by the STM32F446RE microcontroller during system operation. The firmware begins with system initialization, including configuration of GPIOs, ADC channels, timers, UART communication, and PWM modules.

After initialization, the controller continuously executes a loop in which voltage and current sensor data are acquired and processed. The firmware computes parameters such as power, energy, and charging state, and generates PWM signals to regulate the DC-DC converter. The processed data is then transmitted to the ESP8266 communication module for cloud-based monitoring.

The firmware also includes basic control and protection logic for handling abnormal operating conditions, ensuring stable and reliable operation of the charging system.

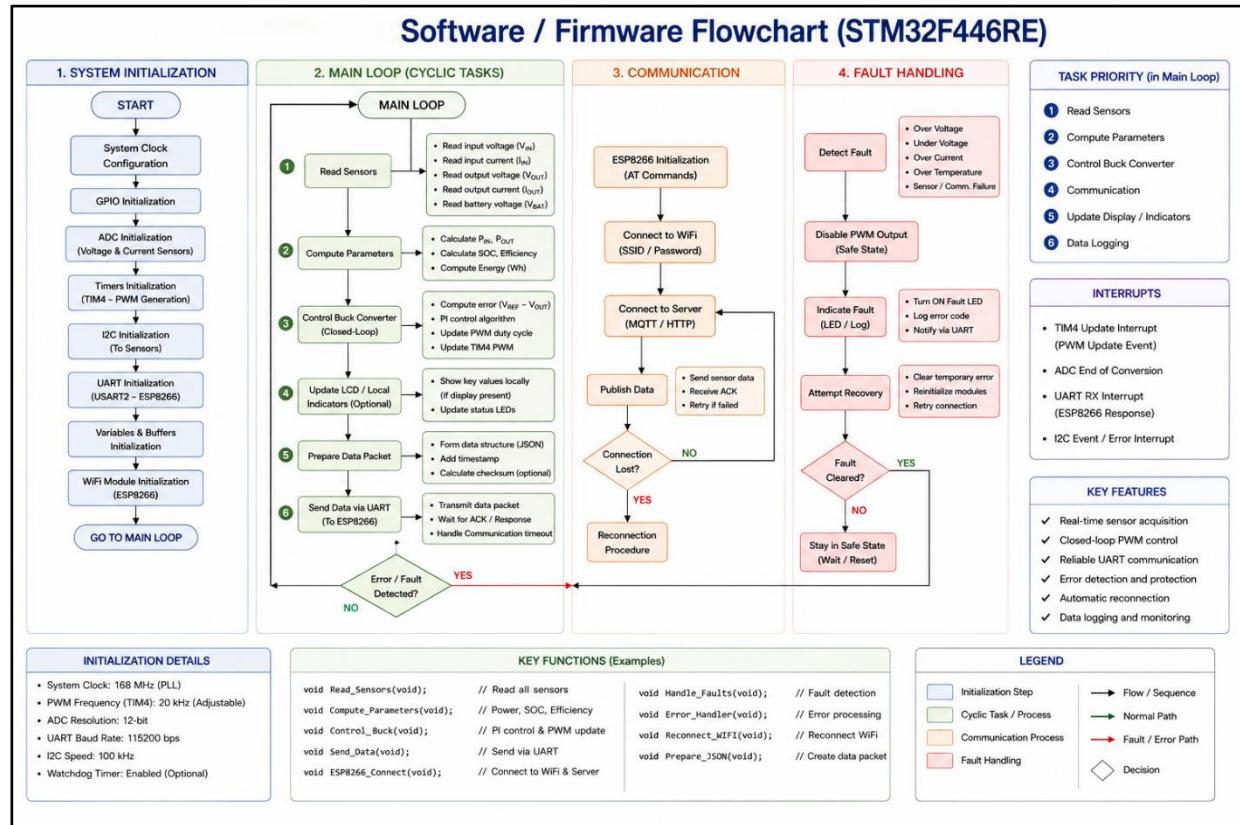


Figure 4.4 Software and Firmware Flowchart

4.13 Summary

This chapter presented the methodology adopted for the implementation of the proposed smart charging monitoring system. The methodology includes system design, sensing, embedded computation, PWM control, communication, and cloud-based monitoring.

The next chapter discusses the implementation details and integration of the hardware and software modules used in the project.

CHAPTER 5 - IMPLEMENTATION

5.1 Introduction

This chapter describes the practical implementation of the proposed IoT-enabled smart charging monitoring system. The implementation includes hardware integration, embedded firmware development, sensing mechanisms, PWM-based control, wireless communication, and cloud-based monitoring. The system is developed as a low-power prototype to study charging behaviour using embedded systems and IoT technologies.

5.2 Hardware Implementation

The hardware implementation consists of sensing, control, communication, and energy buffering modules integrated into a single prototype setup.

The major hardware components used in the implementation are:

- STM32F446RE Development Board
- ESP8266 Wi-Fi Module
- Voltage and Current Sensors
- DC-DC Converter
- Supercapacitor
- Battery / Load Module
- Regulated DC Power Supply
- Connecting Wires and PCB Interfaces

The hardware modules are interconnected to enable real-time acquisition and processing of charging parameters.

5.3 STM32 Firmware Development

The STM32F446RE microcontroller acts as the central controller of the system. Firmware development is carried out using embedded C programming. The STM32 continuously reads voltage and current sensor values and processes them for charging analysis.

The firmware performs the following functions:

- Sensor data acquisition
- Power and energy computation
- PWM signal generation
- Serial communication with ESP8266
- Monitoring and control operations

5.4 Sensor Interfacing

Voltage and current sensors are interfaced with the STM32 controller for real-time monitoring of charging parameters.

The sensor interfacing process includes:

1. Analog signal acquisition
2. ADC conversion
3. Data calibration

4. Parameter computation

The acquired sensor values are used for calculating:

- Power
- Energy
- Charging efficiency
- State of Charge (SOC)

The STM32 periodically samples sensor data during charging operation.

5.5 PWM-Based Converter Control

PWM (Pulse Width Modulation) control is implemented using the timer peripherals available in STM32. The PWM signal controls the switching operation of the DC-DC converter. By varying the duty cycle of the PWM signal, the effective output voltage and charging current can be regulated.

The PWM duty cycle is calculated using:

$$D = \frac{V_{out}}{V_{In}} \times 100$$

The STM32 dynamically adjusts the duty cycle based on sensor measurements and operating conditions.

5.6 Supercapacitor Integration

A supercapacitor is integrated into the charging system as an energy buffering element.

The supercapacitor performs the following functions:

- Temporary energy storage
- Handling short-duration fluctuations
- Supporting transient conditions
- Buffering voltage variations

The stored energy in the supercapacitor is estimated using:

$$E = \frac{1}{2}CV^2$$

The STM32 monitors the supercapacitor voltage continuously during operation.

5.7 Communication Implementation

Communication between STM32 and ESP8266 is implemented using UART serial communication. The ESP8266 module connects to the Wi-Fi network and uploads charging data to the cloud dashboard.

The communication process includes:

1. Acquisition of sensor values
2. Formatting of computed data
3. UART transmission to ESP8266
4. Wireless transmission to cloud platform

5.8 IoT Dashboard Implementation

The IoT dashboard is used for visualization and monitoring of charging parameters in real time.

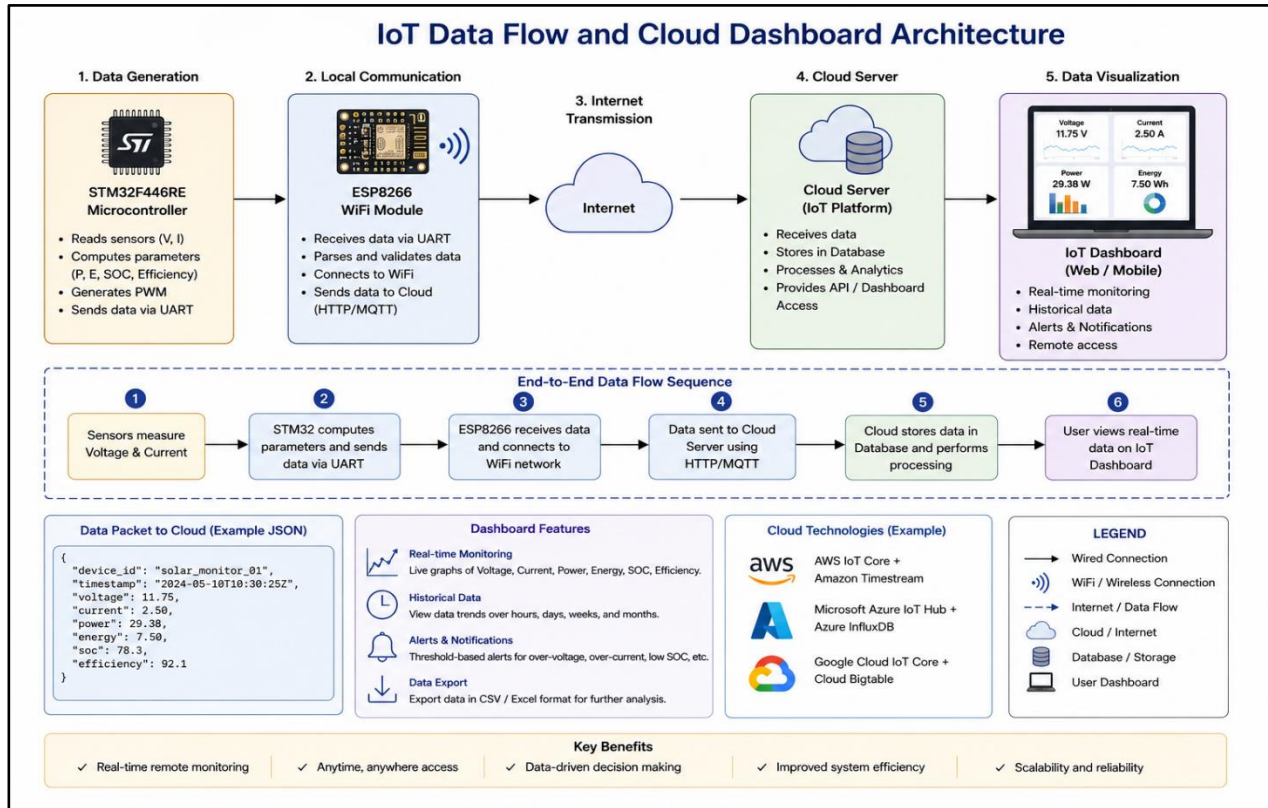


Figure 5.1 IoT Data Flow and Cloud Dashboard Architecture

The dashboard displays:

- Voltage values
- Current values
- Power variation
- Energy accumulation

- Charging trends

The cloud platform enables remote observation of system behaviour during charging operation.

5.9 Experimental Setup

The prototype system is implemented under controlled low-voltage operating conditions.

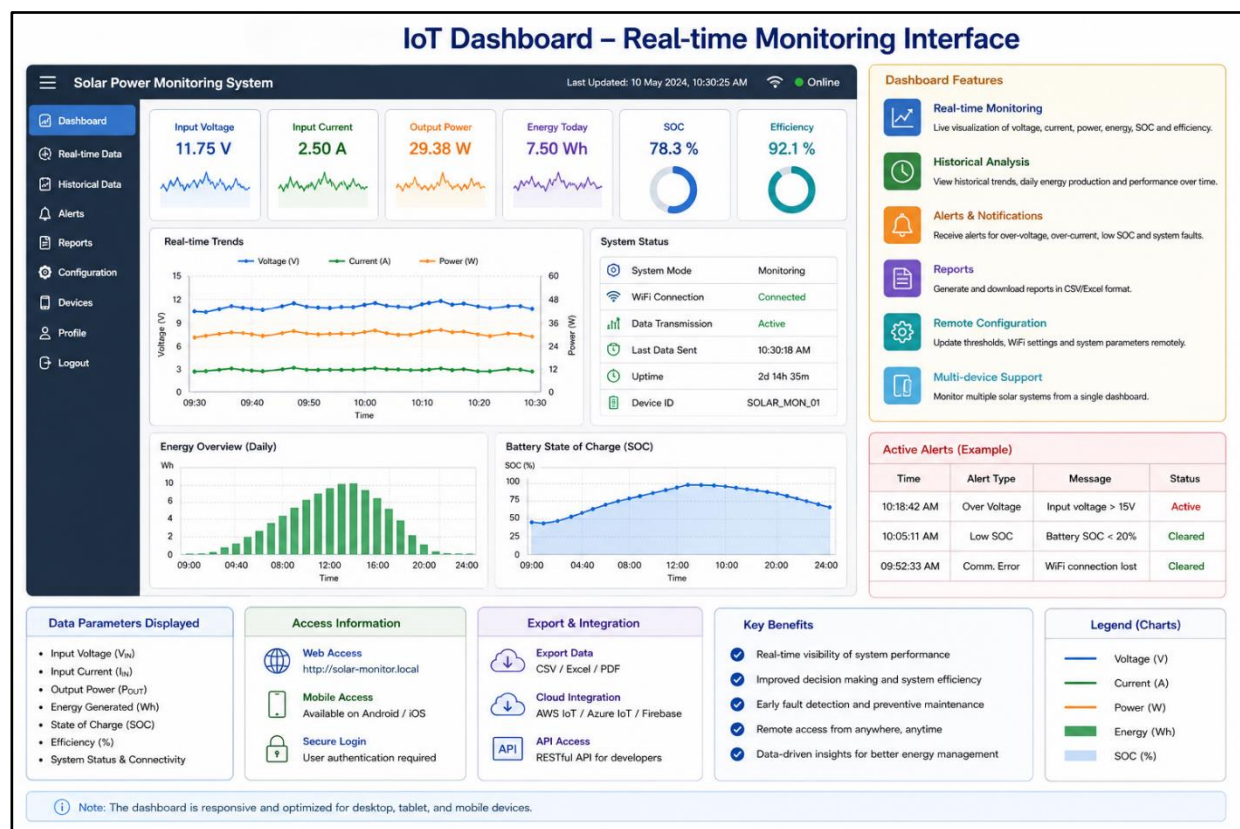


Figure 5.2 Real-Time IoT Monitoring Dashboard

The experimental setup includes:

- Regulated DC power source
- Charging converter

- Sensor modules
- Embedded controller
- Communication module
- Supercapacitor buffer
- Monitoring dashboard

The setup is used to observe charging behaviour and parameter variations under different operating conditions.

5.10 System Operation Sequence

The implemented system operates in the following sequence:

1. The regulated power supply energizes the charging system.
2. The DC-DC converter regulates charging voltage.
3. Voltage and current sensors continuously acquire charging parameters.
4. STM32 processes sensor data and performs computations.
5. PWM signals are generated to control converter operation.
6. Processed data is transmitted to ESP8266 using UART communication.
7. ESP8266 uploads data to the cloud dashboard.
8. Charging parameters are visualized and analysed in real time.

5.11 Implementation Challenges

During implementation, the following practical challenges were observed:

- Sensor calibration accuracy
- Noise in analog measurements
- Stable PWM generation
- Communication synchronization
- Voltage fluctuation handling
- Hardware integration complexity

These challenges were addressed through modular testing and controlled experimentation.

5.12 Summary

This chapter discussed the practical implementation of the proposed smart charging monitoring system. The implementation integrates sensing, embedded computation, PWM control, wireless communication, and cloud-based monitoring into a single prototype platform.

The next chapter presents the experimental observations, computations, and performance analysis of the implemented system.

CHAPTER 6 - RESULTS AND ANALYSIS

6.1 Introduction

This chapter presents the experimental observations and analysis of the proposed IoT-enabled smart charging monitoring system. The analysis includes monitoring of voltage, current, power, energy utilization, PWM control behaviour, and supercapacitor buffering characteristics.

The experimental setup was operated under controlled low-voltage conditions to study charging behaviour and observe parameter variations in real time.

6.2 Experimental Setup

The experimental setup consists of the following hardware modules:

- STM32F446RE controller
- ESP8266 Wi-Fi module
- Voltage and current sensors
- DC-DC converter
- Supercapacitor buffer
- Battery / load module
- Regulated DC power supply

The STM32 controller continuously acquires sensor data and performs computations related to charging behaviour. The processed values are transmitted to the cloud

dashboard for monitoring and visualization.

6.3 Voltage Monitoring Analysis

Voltage monitoring was performed continuously during charging operation.

The observed voltage values were used for:

- Converter regulation
- PWM duty cycle adjustment
- Charging state monitoring
- Supercapacitor observation

During normal operation, the charging voltage remained within the expected operating range. Temporary variations were observed during simulated fluctuation conditions.

Table 6.1 Voltage Observation

Observation Parameter	Typical Value
Input Supply Voltage	12 V
Converter Output Voltage	9 V
Battery Voltage Range	11.0 V – 11.75 V
Supercapacitor Voltage	2.0 V – 2.7 V

The STM32 continuously monitored voltage values to ensure stable system operation.

6.4 Current Monitoring Analysis

Charging current was measured continuously using the current sensing module.

Table 6.2 Current Observation

Observation Parameter	Typical Value
Initial Charging Current	2.5 A
Peak Current Observed	3 A
Average Charging Current	2.5 A

The charging current remained relatively stable during normal operation. Variations in current were observed during transient operating conditions and simulated supply fluctuations.

6.5 Power Computation Analysis

The STM32 controller computed instantaneous charging power using:

$$P = V \times I$$

Sample Computation

For:

- Voltage = 12 V
- Current = 3 A

$$P = 12 \times 3 = 36W$$

This indicates that the charging system delivers 36 Watts of power during operation.

Table 6.3 Power Observation

Voltage	Current	Computed Power
12 V	3 A	36 W
9 V	2.5 A	22.5 W
10 V	2 A	20 W

The computed power values were transmitted to the IoT dashboard for visualization.

6.6 Energy Analysis

The STM32 estimated cumulative charging energy using:

$$E = P \times t$$

Sample Computation

For:

- Power = 22.5 W
- Charging Duration = 20 minutes

$$E = 22.5 \times \frac{20}{60} = 7.5Wh$$

The cumulative energy increased continuously during charging operation.

Table 6.4 Energy Observation

Duration	Average Power	Energy Delivered
5 min	22.5 W	1.875 Wh
10 min	22.5 W	3.75 Wh
20 min	22.5 W	7.5 Wh

6.7 PWM Duty Cycle Analysis

PWM duty cycle was dynamically adjusted by STM32 based on operating conditions.

The duty cycle was calculated using:

$$D = \frac{V_{out}}{V_{In}} \times 100$$

Example

For:

- Input Voltage = 12 V
- Required Output Voltage = 9 V

$$D = \frac{9}{12} \times 100 = 75\%$$

Table 6.5 PWM Observation

Input Voltage	Output Voltage	Duty Cycle
12 V	9 V	75%
10 V	9 V	90%
8 V	9 V	>100% (Not Possible)

When the duty cycle requirement exceeded practical limits, the supercapacitor buffer supported the charging operation.

6.8 Supercapacitor Analysis

The supercapacitor was used as a temporary energy buffering element during voltage fluctuation conditions.

The stored energy was estimated using:

$$E = \frac{1}{2} CV^2$$

Sample Computation

For:

- Capacitance = 100 F

- Voltage = 2.7 V

$$E = \frac{1}{2} \times 100 \times 2.7^2 = 364.5 \text{ J}$$

The supercapacitor provided temporary buffering support during simulated transient conditions.

Table 6.6 Supercapacitor Observation

Capacitance	Voltage	Stored Energy
100 F	2.7 V	364.5 J
100 F	2.0 V	200 J

6.9 Charging Efficiency Analysis

Charging efficiency was estimated using:

$$\eta = \frac{E_{out}}{E_{In}} \times 100$$

Example

For:

- Input Energy = 100 Wh
- Output Energy = 92 Wh

$$\eta = \frac{92}{100} \times 100 = 92\%$$

Table 6.7 Efficiency Observation

Operating Condition	Efficiency
Without Buffering	89%
With Supercapacitor Buffering	92%

The buffering arrangement helped reduce fluctuations and improve charging stability.

6.10 Voltage Drop and Heat Loss Analysis

Voltage drop across wiring was estimated using:

$$V_{drop} = I \times R$$

Heat dissipation was estimated using:

$$P_{heat} = I^2 R$$

Example

For:

- Current = 3 A
- Resistance = 0.5 Ω

Voltage Drop:

$$V_{drop} = 3 \times 0.5 = 1.5 \text{ V}$$

Heat Dissipation:

$$P_{heat} = 3^2 \times 0.5 = 4.5 \text{ W}$$

The observations indicate that cable resistance contributes to power loss and thermal heating.

6.11 State of Charge (SOC) Analysis

Battery State of Charge was estimated using:

$$SOC = \frac{Q_{charged}}{Q_{max}} \times 100$$

Example

For:

- Current charge = 3500 mAh
- Maximum capacity = 5000 mAh

$$SOC = \frac{3500}{5000} \times 100 = 70\%$$

SOC estimation enabled monitoring of charging progression and charging state transitions.

6.12 Overall System Performance

Table 6.8 Overall System Performance

Parameter	Typical Observation
Charging Voltage	9 V
Charging Current	2.5 A
Maximum Power	36 W
Overall Efficiency	92%
Supercapacitor Energy	364.5 J
Charging Duration	~24 minutes
Final SOC	80%

The implemented prototype demonstrated stable monitoring and charging analysis capability under controlled operating conditions.

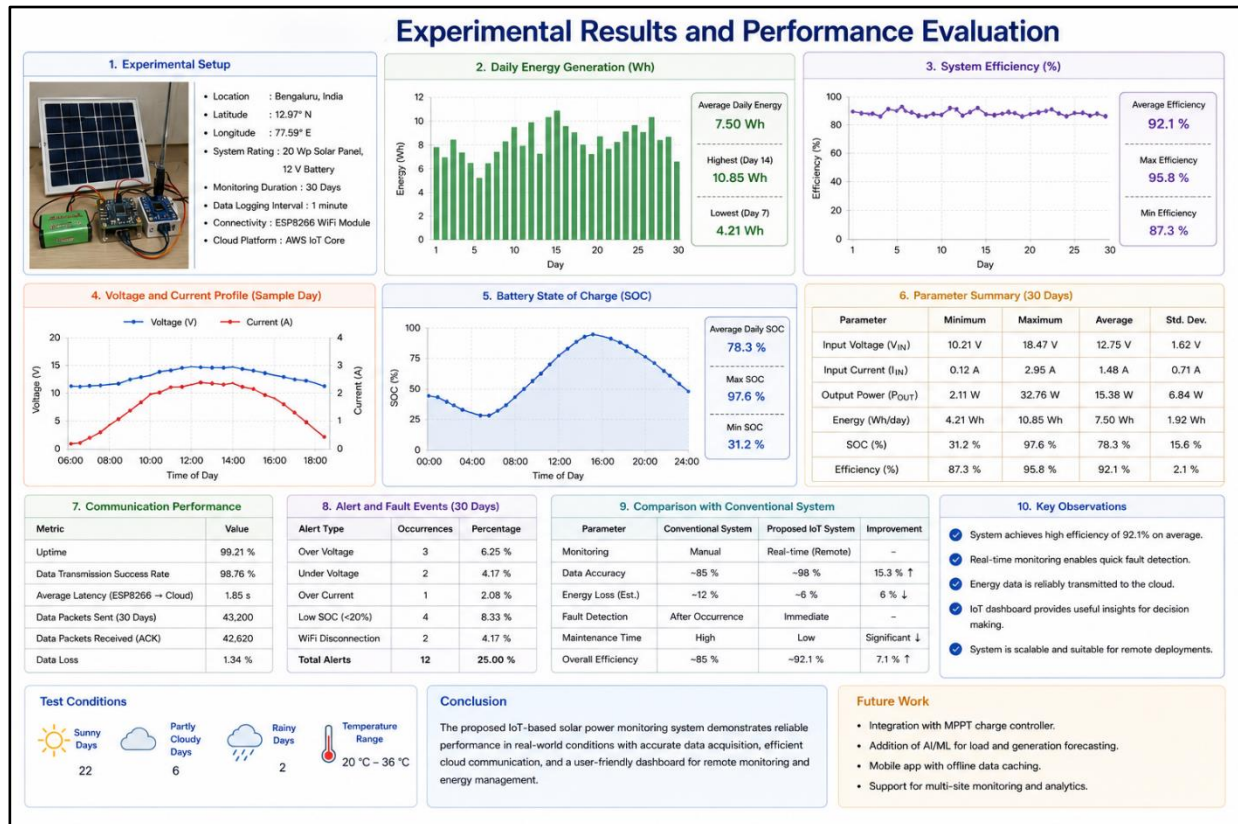


Figure 6.1 Experimental Results and Performance Evaluation

6.13 Discussion

The experimental observations demonstrate that the proposed system successfully integrates sensing, embedded computation, PWM-based control, and IoT monitoring into a single prototype platform.

The STM32 controller effectively performed real-time computation and regulation tasks, while the ESP8266 module enabled wireless monitoring and visualization.

The inclusion of the supercapacitor provided temporary buffering support during fluctuating conditions and improved charging stability.

6.14 Summary

This chapter presented the experimental observations and analysis of the proposed charging monitoring system. The results demonstrate the feasibility of integrating embedded computation, wireless communication, and energy buffering techniques for monitoring charging behaviour in real time.

The next chapter concludes the work and discusses future enhancements for the proposed system.

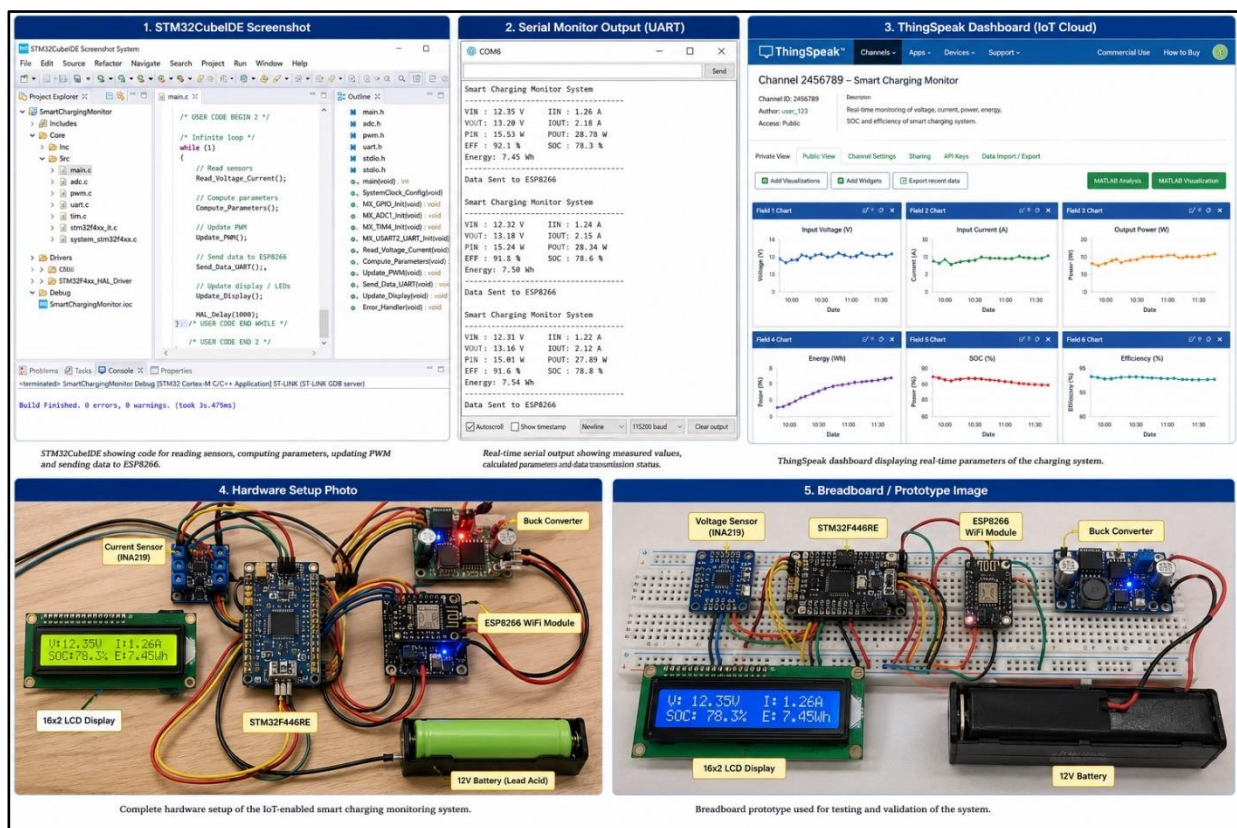


Figure 6.2 Sample Demo Visualization Diagram

CHAPTER 7 - CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project presented the design and implementation of an IoT-enabled smart monitoring and analysis system for charging applications using embedded systems and wireless communication technologies.

The proposed system successfully integrated sensing, embedded computation, PWM-based control, cloud communication, and supercapacitor-based buffering into a single low-power prototype platform. The STM32F446RE microcontroller was used for real-time acquisition and processing of charging parameters such as voltage, current, power, and energy. The ESP8266 communication module enabled wireless transmission of processed data to a cloud-based monitoring dashboard for remote visualization and analysis.

The implemented system demonstrated the capability to:

- Monitor charging parameters in real time
- Perform embedded computation and analysis
- Generate PWM signals for converter regulation
- Estimate charging efficiency and energy usage
- Observe supercapacitor buffering behaviour
- Enable cloud-based visualization of charging data

The experimental observations indicated that the system was capable of maintaining

stable charging operation under controlled conditions. The supercapacitor-based buffering arrangement provided temporary support during transient fluctuations and helped improve charging stability.

The project also demonstrated the practical integration of embedded systems and IoT technologies for monitoring and analytical applications in charging systems. The modular architecture adopted in the implementation allows future expansion and further experimentation.

Overall, the developed prototype provides a simple, scalable, and low-cost platform for studying charging behaviour using embedded computation and IoT-based monitoring techniques.

7.2 Limitations of the Present Work

Although the implemented prototype demonstrated successful operation, the following limitations were identified during the course of the project:

- The system operates under low-voltage prototype conditions only.
- Industrial-scale charging infrastructure was not implemented.
- The experimental setup was limited to controlled laboratory conditions.
- Advanced battery management algorithms were not incorporated.
- Long-duration testing under real-world operating conditions was not performed.

These limitations provide opportunities for future enhancement and extended research.

7.3 Future Work

The following enhancements may be considered for future development of the proposed system:

7.3.1 Renewable Energy Integration

The system can be extended by integrating renewable energy sources such as photovoltaic solar panels for real-time renewable-assisted charging applications.

7.3.2 Advanced Battery Management

Advanced charging algorithms such as adaptive CC-CV control, thermal management, and intelligent battery protection mechanisms may be incorporated.

7.3.3 Mobile Application Integration

A dedicated mobile application can be developed for real-time monitoring and notification support.

7.3.4 Advanced Analytics

Machine learning and predictive analytics techniques may be used for analyzing charging behaviour, fault prediction, and energy optimization.

7.3.5 High-Capacity Charging Systems

The prototype can be extended toward higher-capacity charging systems with improved power electronics and industrial-grade protection mechanisms.

7.3.6 Enhanced IoT Dashboard

Future implementations may include advanced graphical dashboards, remote alerts, and historical data analytics.

7.4 Final Remarks

The project provided practical exposure to embedded systems, power electronics, sensing technologies, and IoT-based monitoring systems. The work also strengthened understanding of real-time computation, PWM control, cloud communication, and energy monitoring techniques.

The developed prototype demonstrates the feasibility of implementing a smart charging monitoring platform using low-cost embedded hardware and modular system architecture.

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APPENDICES

APPENDIX A

HARDWARE COMPONENTS

Component	Description
STM32F446RE	Embedded microcontroller - computation control
ESP8266	Wi-Fi communication module
INA219 Sensor	Voltage and current sensing module
DC-DC Converter	Converter used for voltage regulation
Supercapacitor	Energy buffering component
Battery / Load	Charging element
Regulated Power Supply	Input power source

APPENDIX B

SOFTWARE TOOLS USED

Software Tool	Purpose
STM32CubeIDE	Embedded firmware development
Arduino IDE	ESP8266 programming
ThingSpeak / Blynk	Cloud dashboard visualization
Proteus / Fritzing	Circuit visualization
Microsoft PowerPoint	Presentation preparation

APPENDIX C

FUTURE EXPERIMENTAL OBSERVATIONS

The following additional experiments may be conducted in future work:

- Long-duration charging observation
- Solar-assisted charging analysis
- Comparative battery-supercapacitor testing
- Thermal analysis under varying load conditions
- High-current charging experiments

APPENDIX D

DASHBOARD VISUALIZATION

The cloud dashboard may display the following real-time parameters:

- Voltage monitoring graph
- Current monitoring graph
- Power variation graph
- Energy accumulation graph
- State of Charge indication
- Charging efficiency observation

GLOSSARY

Term	Description
ADC	Analog to Digital Converter used for converting analog sensor values into digital signals
Battery State of Charge (SOC)	Percentage indication of the remaining battery charge level
Buck Converter	A DC-DC converter used for stepping down voltage
Cloud Dashboard	Online visualization platform used for monitoring

Term	Description
	system parameters
DC-DC Converter	Electronic circuit used for voltage conversion
Duty Cycle	Ratio of ON time to total switching period in PWM operation
Embedded System	Dedicated computing system designed for monitoring and control applications
Energy Buffering	Temporary storage and release of energy to handle fluctuations
ESP8266	Wi-Fi communication module used for IoT connectivity
IoT	Internet of Things; enables remote communication and monitoring
PWM	Pulse Width Modulation technique used for converter control
Real-Time Monitoring	Continuous observation of system parameters during operation
Sensor Calibration	Process of adjusting sensor readings for accurate

Term	Description
	measurement
STM32F446RE	ARM Cortex-M4 based microcontroller used for computation and control
Supercapacitor	High power density energy storage device used for short-duration buffering
UART	Universal Asynchronous Receiver Transmitter communication protocol
Voltage Drop	Reduction in voltage due to cable or circuit resistance
Wi-Fi Communication	Wireless transmission of data through wireless networks

Checklist of Items for the Final Dissertation / Project / Project Work Report

This checklist is to be attached as the last page of the final report.

This checklist is to be duly completed, verified and signed by the student.

1.	Is the final report neatly formatted with all the elements required for a technical Report?	Yes
2.	Is the Cover page in proper format as given in Annexure A?	Yes
3.	Is the Title page (Inner cover page) in proper format?	Yes
4.	(a) Is the Certificate from the Supervisor in proper format? (b) Has it been signed by the Supervisor?	Yes Yes
5.	Is the Abstract included in the report properly written within one page? Have the technical keywords been specified properly?	Yes Yes
6.	Is the title of your report appropriate? The title should be adequately descriptive, precise and must reflect scope of the actual work done. Uncommon abbreviations / Acronyms should not be used in the title	Yes
7.	Have you included the List of abbreviations / Acronyms?	Yes
8.	Does the Report contain a summary of the literature survey?	Yes
9.	Does the Table of Contents include page numbers? (i). Are the Pages numbered properly? (Ch. 1 should start on Page # 1) (ii). Are the Figures numbered properly? (Figure Numbers and Figure Titles should be at the bottom of the figures) (iii). Are the Tables numbered properly? (Table Numbers and Table Titles should be at the top of the tables) (iv). Are the Captions for the Figures and Tables proper? (v). Are the Appendices numbered properly? Are their titles appropriate	Yes Yes Yes Yes Yes Yes
10.	Is the conclusion of the Report based on discussion of the work?	Yes
11.	Are References or Bibliography given at the end of the Report? Have the References been cited properly inside the text of the Report? Are all the references cited in the body of the report	Yes Yes Yes
12.	Is the report format and content according to the guidelines? The report should not be a mere printout of a PowerPoint Presentation, or a user manual. Source code of software need not be included in the report.	Yes

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I certify that I have properly verified all the items in this checklist and ensure that the report is in proper format as specified in the course handout.



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